適用於低訊雜比環境之改良式 DEMON 演算法於邊緣運算平台之即時實現

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摘　　要

本研究旨在於嵌入式系統實現具備即時處理能力之 DEMON（Detection of Envelope Modulation On Noise）演算法，建立一套適用於水聲訊號分析之即時處理架構。首先建立完整之 DEMON 訊號處理流程，括帶通濾波、解析訊號建構、Hilbert Transform、包絡擷取、低通濾波及快速傅立葉轉換（Fast Fourier Transform, FFT），並針對各處理步驟進行離散化與演算法分析，經數學模型推導與訊號分析可知，此方法可從高頻載波訊號中萃取低頻調變特徵，經包絡檢測後之頻譜將呈現目標旋轉頻率及其倍頻成分，其特徵頻率可反映船舶推進系統及旋轉機械之運轉資訊，可作為船舶聲紋特徵辨識之重要依據，本研究將此演算法配搭嵌入式系統進行邊緣運算實現。然而，若實海域的原始音檔訊雜比(Signal-to-Noise Ratio, SNR)在 3~4 分貝間，若未先經過有效濾波直接使用 DEMON 演算法其效果不佳，故為提升訊號特徵之辨識能力，本研究進一步針對包絡檢測後之頻譜進行目標訊號增強，並將處理後之頻譜重建為時域訊號，再以固定時間區段進行分割處理。各時間區段經由 DEMON 演算法取得其低頻特徵頻譜後，再透過短時傅立葉轉換（Short-Time Fourier Transform, STFT）進行時頻分析，建立 DEMONgram，以呈現長時間訊號之頻率變化特徵。本研究以實海域歷史水聲資料進行測試，驗證本團隊所建立之改良式 DEMON 演算法流程與訊號處理架構，同步分析不同實際海域所量測之原始 wav 音檔，在不同訊雜比條件下之特徵辨識能力。

此外，為將 DEMON 演算法實現邊緣運算，本研究以 C 語言進行編譯，以 Jetson nano 嵌入式系統作為實作平台，針對記憶體使用、運算量及處理時間進行分析與最佳化，以評估其即時處理之可行性。實驗顯示，在 48kHz 取樣率下之音檔，每 30 秒音訊僅需 2.6 秒處理，速度為實際的 11.3 倍。同步測試 384kHz 高取樣率之音檔，龐大資料量使處理時間拉長至 26.9 秒，速度降至 1.09 倍。對比證實，即使高負載皆能在 30 秒內完成 DEMON 演算法即時運算。嵌入式平台實驗結果驗證本團隊改良之 DEMON 演算法能有效萃取水聲訊號中的低頻調變特徵，無論是面對常規 (48k) 還是極高頻 (384k) 的水下音訊取樣率，系統皆能保證在音訊進來的 30 秒內完成運算解析與 DEMONgram 成像，以確保實務即時監測不會產生運算延遲，在頻譜

解析度與運算效率之間取得適當平衡，證實目前具備於嵌入式系統進行即時水聲訊號運算及分析之可行性。研究成果可作為後續即時船舶聲紋辨識及海上水聲監測系統之基礎。

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關鍵詞：DEMON、Hilbert Transform、嵌入式系統、聲紋訊號處理

(音檔來源：淡水河音檔，每段音檔約為30min)

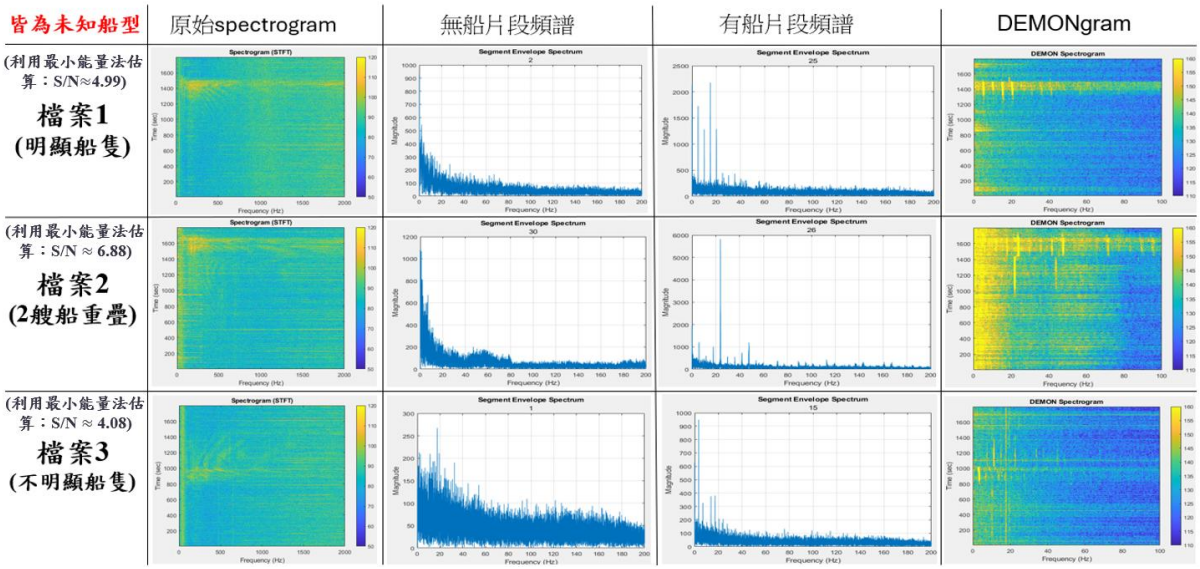


圖 1.(左)原始 spectrogram；(中左)經改良演算法對無船片段無法看出倍頻結構；(中右) 經改良演算法對有船片段能明顯看出倍頻結構；(右)經改良演算法呈現長時間訊號之頻率變化特徵

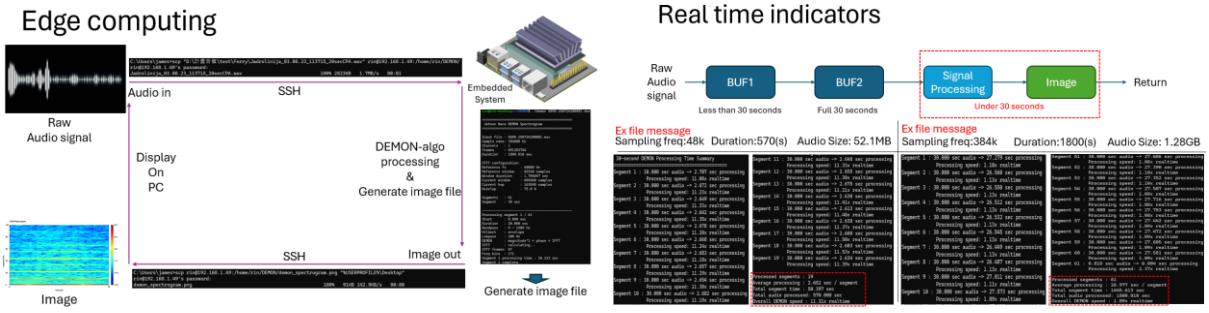


圖 2. (左)邊緣計算架構-聲音訊號透過終端機進入嵌入式系統經計算產生圖檔再由電腦顯示；(右) 於嵌入式系統終端機回傳各段處理時間以及是否在即時處理的預期

Real-Time Implementation of an Improved DEMON Algorithm for Low-SNR Environments on an Edge Computing Platform

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ABSTRACT

This study aims to implement a real-time DEMON (Detection of Envelope Modulation On Noise) algorithm on an embedded system and establish a real-time processing architecture for underwater acoustic signal analysis. First, a complete DEMON signal processing pipeline is established, including band-pass filtering, analytic signal construction, Hilbert transform, envelope extraction, low-pass filtering, and Fast Fourier Transform (FFT). Each processing stage is discretized and analyzed to facilitate implementation on an embedded system. Based on mathematical derivations and signal analysis, the proposed method can extract low-frequency modulation features from high-frequency carrier signals. After envelope detection, the resulting spectrum reveals the target rotational frequency and its harmonic components, which can reflect the operating characteristics of ship propulsion systems and rotating machinery and serve as important features for ship acoustic signature identification. The proposed algorithm is implemented on an embedded system for edge computing. However, when the Signal-to-Noise Ratio (SNR) of raw recordings collected in real marine environments is approximately 3–4 dB, directly applying the DEMON algorithm without effective pre-processing and filtering results in poor performance. Therefore, to improve the identification of signal characteristics, this study further enhances the target signal in the spectrum after envelope detection, reconstructs the enhanced spectrum into a time-domain signal, and then divides the signal into fixed-length segments. Each segment is processed using the DEMON algorithm to obtain its low-frequency feature spectrum, followed by Short-Time Fourier Transform (STFT) for time-frequency analysis to construct a DEMONgram, thereby revealing the frequency variations of long-duration acoustic signals. Historical underwater acoustic recordings collected from actual marine environments are used to evaluate the proposed improved DEMON algorithm and signal processing architecture. Recordings acquired from different marine environments are also analyzed to evaluate feature identification performance under various SNR conditions.

Furthermore, to implement the DEMON algorithm on an edge computing platform, the proposed algorithm is developed in the C programming language and compiled for execution on a Jetson Nano embedded system. Memory usage, computational load, and processing time are analyzed and optimized to evaluate the feasibility of real-time processing. Experimental results show that, for audio recordings sampled at 48 kHz, each 30-second segment requires only 2.6 seconds to process, achieving a real-time processing factor of 11.3×. For recordings with a high sampling rate of 384 kHz, the substantially increased data volume extends the processing time to 26.9 seconds, corresponding to a real-time factor of 1.09×. These results demonstrate that even under high computational loads, the complete DEMON processing can be completed within the corresponding 30-second audio interval. The embedded-system experiments verify that the proposed improved DEMON algorithm can effectively extract low-frequency modulation features from underwater acoustic signals. For both conventional 48 kHz and high-rate 384 kHz underwater acoustic recordings, the system can complete signal processing, analysis, and DEMONgram generation within 30 seconds of audio acquisition, thereby preventing computational delays in practical real-time monitoring applications. The proposed system achieves an appropriate

balance between spectral resolution and computational efficiency, demonstrating the feasibility of real-time underwater acoustic signal processing and analysis on an embedded edge-computing platform. The results provide a foundation for future real-time ship acoustic signature identification and marine underwater acoustic monitoring systems.

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Keywords: DEMON, Hilbert Transform, Embedded System, Acoustic Signal Processing

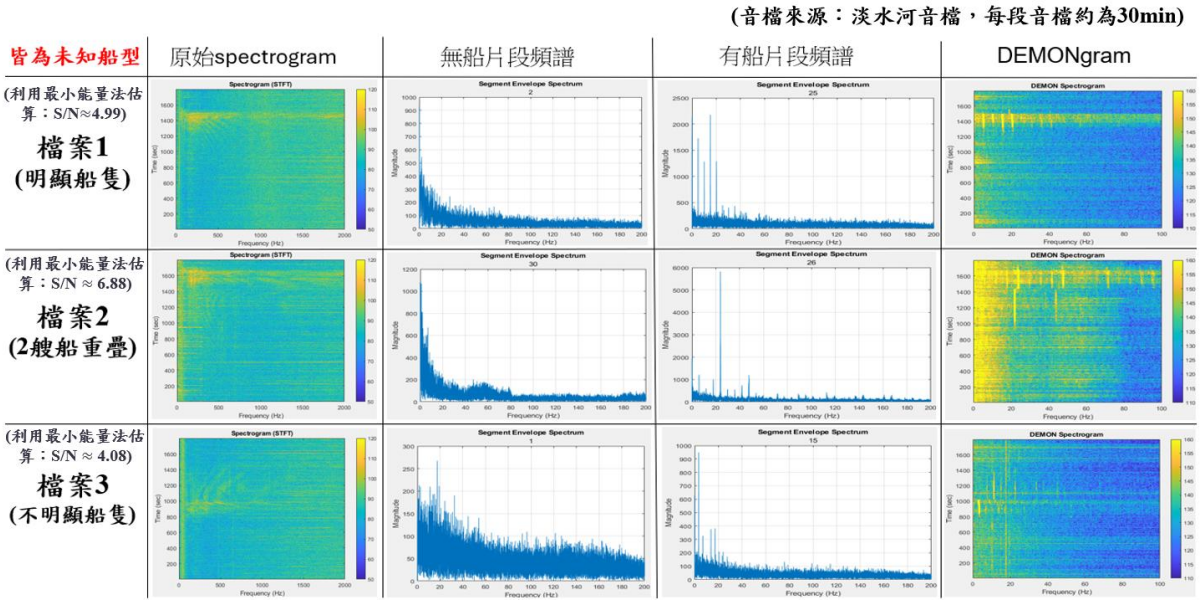


Figure 1. (Left) Original spectrogram; (Middle left) segment without ship signals, showing no distinct harmonic structure after applying the improved algorithm; (Middle right) segment with ship signals, showing clearly identifiable harmonic structures; (Right) long-term frequency variation characteristics obtained using the improved algorithm.

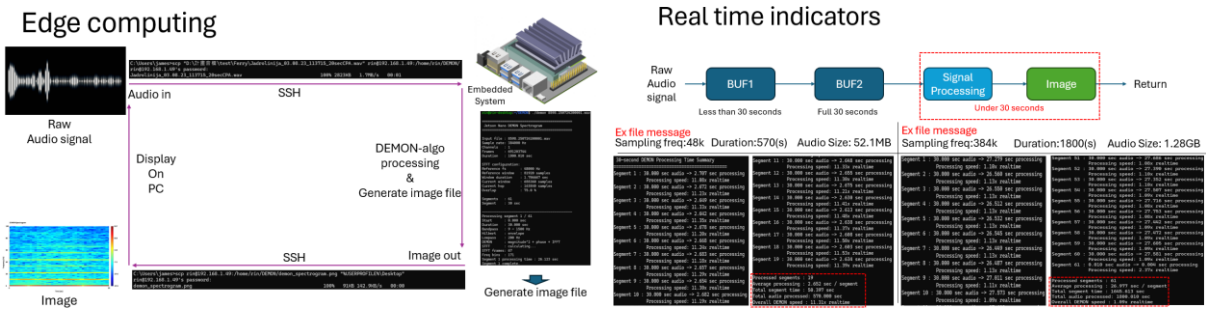


Figure 2. (Left)Edge Computing Architecture - Audio signals are input into the embedded system through the terminal, processed by the DEMON algorithm, and converted into image files for display on a remote computer; (Right) The embedded system terminal reports the processing time of each segment and verifies whether real-time processing requirements are met.